Mathematical overview: facts, assumptions and limits

The nine distinct shape/fill tiles form the affine plane with coordinates (shape,fill) in $\{0,1,2\}^2$. An allowed three uses different tiles and has each attribute independently all equal or all different. Any pair has unique completion -x-y modulo 3; the plane has 12 triples and maximum triple-free size 4. Physical alignment in a straight row is irrelevant: there are four wrap-around triples not in ordinary tic-tac-toe. Ownership counters are an extra label and do not change the fill attribute.

Private-collection avoidance game: the first player can force a win. First claim any center c, then respond to each second-player tile x with its unique opposite 2c-x. The other eight tiles split into four pairs. First never owns an opposite pair about c, so cannot have a triple through c. Any first-owned triple avoiding c would reflect to an already second-owned triple. Thus the first safely responds; second must lose by its fourth claim, because nine total claims would otherwise require one player to own a triple. This is a new game: in the base shared-collection adding game every legal play lasts four turns and the second player wins. Do not transfer that result to this game.

Two ownership colors cannot cover all nine tiles without a monochromatic allowed three, because each triple-free class has at most four tiles. Three ownership colors suffice; a checked witness appears below. With a number attribute taking three values, a pair still has unique completion. A nine-tile triple-free choice exists with one numbered copy of every shape/fill position; no maximum theorem for all 27 tiles is required here.

Children can experiment with claim choices and coloring, then explain a guarantee or obstruction. Grades 2–3 can play P1 and design P2 with a partner enforcing triple checks. Grades 4–5 can justify the strategy, the two-color impossibility, and the number-layer design. These pages do not offer a forced K–1 adaptation: reliable independent two-attribute checking is essential. Arithmetic is counting to nine, recognizing equality/difference, and optionally multiplication; modular notation is for adults only.

Materials, route and launch

Per pair: three reusable nine-tile sets (27 cards), number stickers 1/2/3 so each shape/fill combination appears once at each number, 12 ownership counters each of red/blue/green (36), one large nine-site mat, one folder/tray, pencil and two tracing sheets. The student sites are 108 pt (38.1 mm); the layer sites are 58 pt (20.5 mm). Use the large mat as the tile display and put ownership counters up to 15 mm across near a cell corner, keeping the printed shape and fill visible. Existing base cards about 45 mm across stay beside the mat or on the table; they are larger than its 38.1 mm cells. For P3, choose the physical numbered cards on the table and write the chosen numbers on the two small records. The 58 pt numbered-layer diagrams identify the available copies; neither those diagrams nor the records are placement boards or exact-fit cutting templates. For six pair-sized kits across KK11 / 3333 / 445: 162 numbered cards, 72 counters of each ownership color, six mats/trays and twelve tracing sheets. Only three active older-table kits are essential if KK11 stays with previously accepted Week 36 readiness material; six kits are the full all-table stock, not a recommendation that K–1 attempt these investigations. Three anchored adults remain appropriate. Preparing 27 paper cards per active pair from existing tile sets is realistic; mark numbers before class, and do not rely on children to recopy the full deck.

Let children handle the tiles. As a whole-group demonstration, choose open circle, striped triangle and solid square, and apply both attribute tests separately. Move the cards into a bent arrangement; the triple stays allowed. Then place ownership counters, including a near miss whose three shapes are different but fills are two alike. Before proceeding, each child should check one allowed triple and one near miss without the adult operating the test. Show claiming one tile and keeping private ownership; avoid revealing the optimal first response. Before P3, use the student visual: open circle, striped triangle, solid square pass shape/fill tests. Numbers 1,1,2 fail the number test, so the triple is not allowed; numbers 1,2,3 pass it, so the triple is allowed. These are convention examples, not part of the nine-card target design.

A first return visit can devote 20–30 minutes to repeated P1 play and 15–25 to P2. P3 deserves a separate 20–40-minute visit when the two-attribute rule is secure. Stop after a defensible design or

strategy; completion is not the pacing criterion. Timing and child readiness remain unpiloted. Physical counter/card fit and transfer have not been pretested.

Solutions and hints

P1 / p. 1, private-collection avoid-a-three. First-player win with the center-and-opposites strategy stated above. For c = open circle, the pairs are striped-circle/solid-circle, open-triangle/open-square, striped-triangle/solid-square, solid-triangle/striped-square. Check an opposite pair by putting c beside it and applying the attribute rule. A response cannot have been previously claimed: it belongs to the pair just opened by the second player's move, and all previously opened pairs contain one tile of each owner. The first player owns c plus at most one point from each pair. Every triple not through c reflects to another allowed triple; the second player would have lost before the first player completed that triple. A losing move ends the game immediately; players do not continue filling the mat. The exact minimax solver independently confirms a first win and exhausts every second choice under every possible first center (3,024 losing terminal branches, latest loss on second's fourth claim). Hints, if wanted: look at the four allowed threes through the first claim; ask whether the remaining tiles can be paired. Do not print or dictate the response rule before experimentation. Upper continuation: change the losing condition to a shared three and explain why its winner changes; this is comparison with the base game, not a fourth new investigation.

P2 / p. 2, ownership coloring. A witness, in row order circle/triangle/square and column order open/striped/solid, is:

R R B

R R B

B B G

Every one of the 12 allowed threes has at least two ownership colors. There are 3,762 valid assignments with three named colors; this exhaustive number is adult check evidence, not a child catalog target. Two colors fail by $9 > 4+4$. To justify the bound 4 concretely, use the base five-tile pairing argument: five triple-free chosen tiles would give 10 pairs whose third tiles lie among four unchosen positions; each unchosen tile completes at most two disjoint chosen pairs, a capacity of 8. The argument assumes no duplicates. Hint: first keep each color's chosen group free of triples, then see what is left. Extension: require all color classes to have three cards, or require specified cards to share an owner; verify any proposed arrangement directly rather than infer it from the 4/4/1 witness.

P3 / p. 3, numbered layers. One number matrix in the same shape/fill order is:

1 2 2

2 3 3

2 3 3

Keep the indicated numbered version of each of the nine positions. This uses all numbers and contains no allowed three. To explain without code, examine the 12 possible shape/fill threes: their number triples are never all equal or all different. For an adult general argument, use number-1 = x^2+y^2 modulo 3. Along a nonconstant affine line, the quadratic coefficient is $d_1^2+d_2^2$, which is 1 or 2, never zero for a nonzero direction; the three values are neither constant nor all distinct. This argument is adult context, not a prerequisite. Adding the same amount modulo three to every number produces new witnesses, as does swapping the names of values in the attributes; check whether a child regards those as genuinely different designs. Changing all numbers to 1 makes all 12 shape/fill threes reappear. The task asks for an achievable design, not the maximum of the 27-tile deck. Hint: reuse the saved base collection of all twelve shape/fill triples to pool checking across pairs, rather than ask one child to rediscover that catalog while choosing numbers. Save a nine-position number record and test a suspected three using the physical cards; a second pair can test a design rather than recreate a full catalog. Extension: a nine-card set with two copies of one shape/fill tile breaks the one-version rule but usually has no allowed three. Two copies of a tile are completed only by its third copy. Ask why the page insists on one version of each tile.

Provenance and verification

P1 changes from one shared collection (base Grades 2–3 P5 / Grades 4–5 P5) to private ownership with a genuinely strategic first-player guarantee. P2 colors every tile and compares two versus three ownership colors, beyond the base maximum cap collection. P3 adds a third attribute and makes a concrete nine-position design, beyond base pair completion and nine-tile incidence. These are original extensions; no downloaded activity is copied. The base Week 36 overview supplies the finite-plane definitions and elementary four-tile bound; the Week 1 encore supplies the familiar-material/new-direction planning model.

student/verify.py independently builds all 12 triples from attribute tests, computes maximum cap size, solves the full private-collection game, checks every branch of the mirror strategy, enumerates all two/three-owner colorings, and checks all triples of the nine-number witness. It uses only standard Python. Portable rebuilding is sh student/build.sh OUTPUT_DIR with ReportLab installed; no repository imports are used. Record actual use separately as W36-B1, B2 or B3. All preparation is unpiloted; physical handling/readiness pretests are unperformed.